# Thibault JUILLARD

thibault.juillard@uni-hamburg.de
https://thibjrd.github.io/


## General information

| | |
|---|---|
| Birth date | The 12th of December, 1996. |
| Nationality | French. |
| ORCID | https://orcid.org/0000-0001-9071-0409 |

## Employment History

- Since 07/2025 — **Postdoc, Universität Hamburg**, Department of Mathematics, Germany. Mentor: Sven Möller.
- 10/2024 – 06/2025 — **Teacher and researcher, École polytechnique**, Centre de Mathématiques Laurent Schwartz, Palaiseau, France. Temporary position during the last year of my Ph.D.
- 02/2022 – 09/2024 — **Doctoral researcher, Université Paris-Saclay,** Institut de mathématiques d'Orsay, France.

## Education

- 02/2022 – 06/2025 — **Ph.D. Mathematics, Université Paris-Saclay,** Orsay, France. Supervisor: Anne Moreau. Dissertation title: *Reduction by stages for affine W-algebras and applications*. Defended on the 13th of June, 2025.
- 09/2020 – 01/2022 — **M.Sc. Mathematics, École polytechnique fédérale de Lausanne**, Switzerland. Dissertation title: *Nilpotent orbits, Slodowy slices and BRST cohomology*.
- 08/2017 – 08/2021 — **M.Sc. Mathematics, École polytechnique,** Palaisseau, France.

## Research Publications

### Publication

1. N. Genra and T. Juillard, "Reduction by stages for finite W-algebras," *Math. Z.*, vol. 308, no. 1, Paper No. 15, 36, 2024, ISSN: 0025-5874,1432-1823. DOI: 10.1007/s00209-024-03567-9.

### Prepublications

1. T. Juillard, "A decomposition theorem for minimal W-algebras of type A," 2026. arXiv: 2611.13239 [math.RT]. URL: https://arxiv.org/abs/2611.13239.
2. N. Genra and T. Juillard, "Reduction by stages for affine W-algebras," Preprint, 2025. arXiv: 2501.04501 [math.RT]. URL: https://arxiv.org/abs/2501.04501.

### Thesis

1. T. Juillard, "Reduction by stages for W-algebras and applications," Ph.D. thesis, Université Paris-Saclay, 2025. URL: https://theses.hal.science/tel-05149024.

## Talks

### Conferences and workshops

04/2026 *Kraft–Procesi rule for W-algebras.* Itinerant workshop of the thematic network for algebra, Tours, France.

02/2026 *Kraft–Procesi rule for W-algebras.* Workshop "Vertex algebras and related topics", Trinity College, Dublin, Ireland.

11/2025 *Reduction by stages for affine W-algebras*. "Collaborative Research Center" retreat, Büsum, Germany.

10/2025 *Reduction by stages for affine W-algebras*. Conference "Collaborations in algebra, representation theory and ethics", École normale supérieure, Lyon, France.

06/2025 *Reduction by stages, a geometric approach*. Conference "Representation Theory XIX", IUC, Dubrovnik, Croatia.

06/2024 *Reduction by stages for W-algebras.* Conference "Vertex algebras, geometric representation theory and quantum groups", CIRM, Luminy, France.

*Chiralisation of reduction by stages.* Conference "Groups and their actions", Levico Terme, Italy.

01/2024 *Reduction by stages for W-algebras.* Winter school in mathematical physics, Les Diablerets, Switzerland.

10/2022 *Reduction by stages for finite W-algebras.* Workshop "Quantum symmetries", Centre de recherche en mathématiques, Montréal, Canada.

### Seminars

11/2026 *A decomposition theorem for minimal W-algebras in type A.* Seminar of the Creutzig group, Friedrich-Alexander-University, Erlangen.

09/2026 *A decomposition theorem for minimal W-algebras in type A.* Algebra seminar, School of Mathematics and Statistics, Glasgow.

04/2026 *Reduction by stages for affine W-algebras.* Seminar of the Hausel group, IST Austria, Vienna.

03/2026 *Kraft–Procesi rule for W-algebras.* Seminar "GAT", Université d'Amiens, France.

*Kraft–Procesi rule for W-algebras.* Université de Reims, France.

*Kraft–Procesi rule for W-algebras.* Seminar "GAAO", Université Clermont Auvergne, Clermont-Ferrand, France.

*Kraft–Procesi rule for W-algebras.* Topology and algebraic geometry seminar, Université d'Angers, France.

05/2025 *Reduction by stages for affine W-algebras, a geometric approach.* Algebraic geometry seminar, Université de Lille, France.

01/2025 *Reduction by stages for affine W-algebras, a geometric approach.* Representation theory seminar, I2M, Luminy, France.

12/2024 *Reduction by stages for affine W-algebras, a geometric approach.* Mathematical physics seminar, Hamburg, Germany.

11/2024 *Reduction by stages: from Slodowy slices to affine W-algebras.* Seminar "Groups, representations and geometry", Paris Cité, France.

10/2024 *Reduction by stages for affine W-algebras.* Algebra and representation theory seminar, Tor Vergata, Rome, Italy.

03/2024 *Reduction by stages for W-algebras.* Seminar "AGATA", Montpellier, France.

01/2024 *Reduction by stages for W-algebras.* Algebra seminar, ICJ, Lyon, France.

## Talks (continued)

### Reading groups

10/2026 — *The chiral de Rham complex.* Workshop "Seminar on Vertex Operator Algebras in Thiessow", Thiessow, Rügen.

01/2025 — *Symplectic geometry and characteristic variety.* Reading group "D-modules", IMO, Orsay, France.

03/2024 — *Examples of GIT constructions: quiver varieties.* Reading group "Good moduli stacks", IMO, Orsay, France.

02/2024 — *Equivariant sheaves and linearisations in GIT.* Reading group "Good moduli stacks", IMO, Orsay, France.

11/2023 — *Geometric Invariant Theory.* Reading group "Good moduli stacks", IMO, Orsay, France.

04/2023 — *Double Poisson vertex algebras.* Worshop "Hall algebras and vertex algebras in enumerative geometry", Skye, Scotland.

01/2023 — *Character variety and tautological classes.* Reading group "P = W Conjecture", IMO, Orsay, France.

## Grants

06/2025 — *Vivaldi program*. Travel grant.
Funding institution: Fondation Mathématique Jacques Hadamard.
Host institution: IUC, Dubrovnik, Croatia.

06/2024 — *Vivaldi program*. Travel grant.
Funding institution: Fondation Mathématique Jacques Hadamard.
Host institution: CIRM, Luminy, France.

06/2023 — *Vivaldi program*. Travel grant.
Funding institution: Fondation Mathématique Jacques Hadamard.
Host institution: IUC, Dubrovnik, Croatia.

10/2022 — *Program for junior scientific visibility*. Travel grant.
Funding institution: Fondation Mathématique Jacques Hadamard.
Host institution: CRM, Montréal, France.

## Teaching

### Supervision of exercise sessions

2024 – 2025 — *Euclidean and Hermitian spaces,* Bachelor 2, Polytechnique, Paris, France.

2023 – 2025 — *Differential geometry,* Bachelor 3, Polytechnique, Paris, France.

2022 – 2023 — *Series of functions, differential equations,* Bachelor 2, Polytechnique, Paris, France.

2021 — *Geometry,* Bachelor 2, EPFL, Lausanne, Switzerland.

*Linear algebra 1,* Bachelor 1, EPFL, Lausanne, Switzerland.

*Analysis 1,* Bachelor 1, EPFL, Lausanne, Switzerland.

2020 — *Group theory,* Bachelor 2, EPFL, Lausanne, Switzerland.

### Supervision of students

2026 – 2027 (1 year) — **Master project, Universität Hamburg**, Germany.
Student: Bailey White. (Co-advisor: Christopher Raymond.)
Dissertation title: *The Feigin–Frenkel centre in exceptional types*.

## Teaching (continued)

2026 (3 months) — **Internship, Universität Hamburg**, Germany.
Intern: Ren Grégoire, Master 1, ENS Paris-Saclay, France.
Dissertation title: *Symmetric invariants of the centraliser of a nilpotent element*.

## Mobility

10/2022 — Thematic program "Quantum symmetries". CRM, Montréal, Canada.
Duration: 4 weeks.

03/2022 – 04/2022 — Thematic program "Vertex and Chiral Algebras". IMPA, Rio de Janeiro, Brasil.
Duration: 6 weeks.

## Mathematical interests

**Representation theory** — Simple Lie algebras, vertex algebras, W-algebras, homological algebra, quantisation.

**Algebraic geometry** — Poisson geometry, algebraic groups, geometric invariant theory, Slodowy slices, nilpotent orbits.

## Miscellaneous Experience

### Event organisation

09/2026 – 08/2027 — **ZMP Junior Seminar,** University of Hamburg and Desy, Germany.
Seminar for young researchers in mathematical physics.

09/2023 – 01/2025 — **RéGA,** IHP, Paris, France.
Seminar for young researchers in algebraic geometry.

09/2022 – 10/2023 — **Popularisation seminar,** IMO, Orsay, France.
Ph.D. student seminar.

### Popularisation

08/2024 — **APEX-Maths,** Resita, Romania.
3-week program to initiate high school students to mathematical research.

July, 2021 – 2024 — **Astromaths,** Ardèche, France.
2-week science camp in mathematics and astronomy for high school students.